The Ethiopia Kiremt 2026 Drought Outlook — reproduced with Rosetta + DeepScale

*A worked walk-through of the CHC / FEWS NET deck "**Ethiopia Likely to Experience Very Poor Rains in 2026**" (Chris Funk, July 2026) — the outlook for the June–September **Kiremt** rains — rebuilt with `rosetta` and `deepscale`.*

What the deck concludes

The deck's headline is a very poor Kiremt 2026, from a **convergence of evidence**: rainfall observed so far, El Niño **analog** years, a **dynamic forecast**, and a West-Pacific **teleconnection** cross-check — all pointing dry.

What this notebook builds — and on which data

`rosetta` for acquisition, `deepscale` for the SMPG logic. The deck's observed layer is **CHIRPS** and its forecast **CHIRPS-GEFS**; we use **ERA5** for the gridded observed maps and **NMME** for the dynamic forecast, because **we were unable to use the 2026 CHIRPS data**, so it cannot supply the June anchor or the July forecast. ERA5 is the standard reanalysis stand-in for the 2026 fields; CHIRPS is used wherever the historical record suffices (§9–10). A correspondence check (§10) measures the substitution.

The substitution is not free: the NMME July forecast is near climatology, so — unlike the deck's drier CHIRPS-GEFS — the *dynamic* leg **softens** rather than sharpens the outlook. We flag it at each stage.

Setup

```python
import warnings; warnings.filterwarnings("ignore")
import numpy as np, pandas as pd, xarray as xr
import matplotlib.pyplot as plt
import geopandas as gpd
from shapely.geometry import Point
import rosetta
import deepscale as ds
import deck_style as deck
from datetime import datetime
from deepscale.registry import get_method

GHA        = [-13.5, 24.5, 20.5, 53.0]      # Greater Horn of Africa (deck monitoring extent)
REGION     = [3.0, 15.0, 33.0, 48.0]        # Ethiopia (SMPG analysis extent)
SEASON     = "JJAS"                          # the Kiremt season
TARGET     = 2026
BASELINE   = (1991, 2020)                    # index (Niño-3.4) anomaly baseline (standard ENSO baseline)
CANDIDATES = list(range(1991, 2026))         # ONE window for climatology / percentile baseline / analog pool
woredas = gpd.read_file("data/eth_adm3_woreda.geojson")
print(f"{len(woredas)} woredas loaded")
```

```
690 woredas loaded
```

1 · Fetch the season

One `fetch()` returns normalized ERA5 monthly precipitation over the Greater Horn; we subset Ethiopia and stack it into one Kiremt season per year. The percentile baseline and the analog pool use a single window (`CANDIDATES`, 1991–2025); the Niño-3.4 index uses the standard 1991–2020 anomaly baseline.

```
era_gha = rosetta.fetch("obs/era5", "precip", hindcast=(1991, 2026), region=GHA, verbose=False)["precip"]
era = era_gha.sel(lat=slice(REGION[0], REGION[1]), lon=slice(REGION[2], REGION[3]))
climatology = ds.seasonal_stack(era, SEASON, years=CANDIDATES)
print("Ethiopia subset:", dict(era.sizes), "| ERA5 latest month:", str(era.time.values[-1])[:7])
```

```
Ethiopia subset: {'time': 426, 'lat': 49, 'lon': 61} | ERA5 latest month: 2026-06
```

1b · The forecast leg — a bias-corrected NMME July field

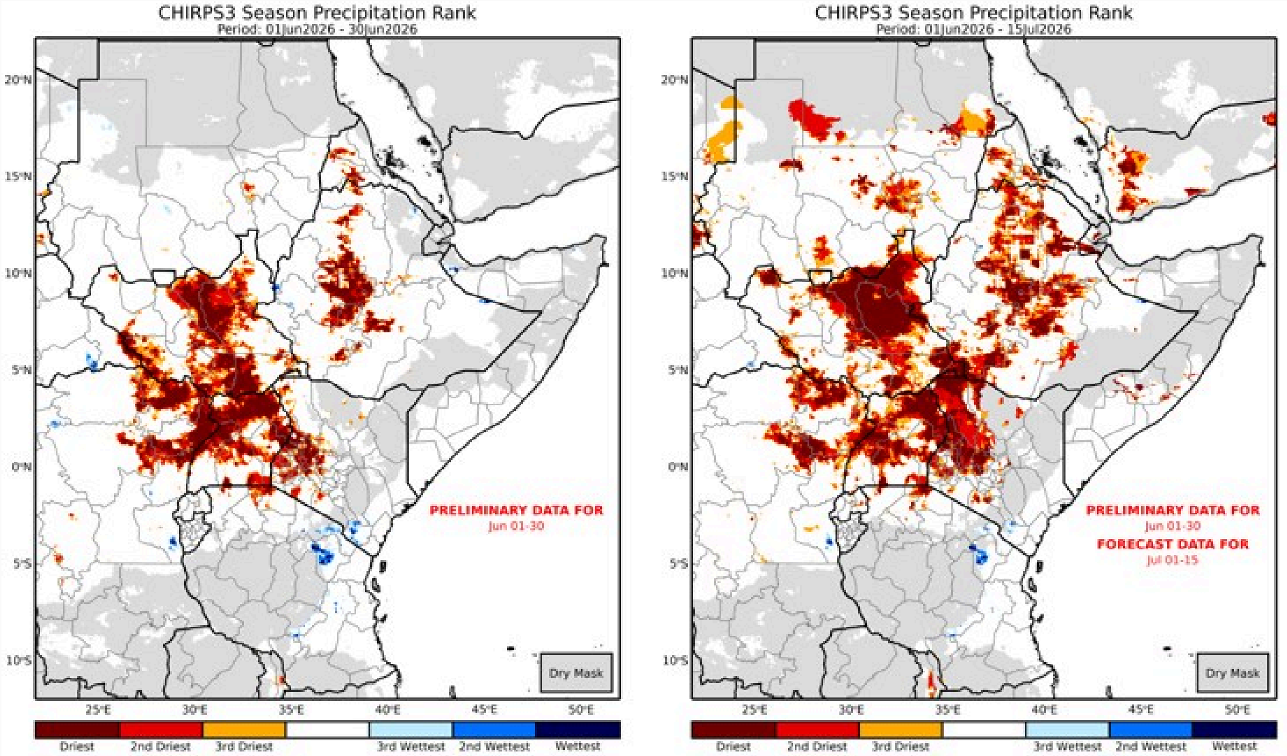
The dynamic July forecast is the **NMME** (three models), gridded-quantile-mapped onto ERA5. We build it over the GHA extent and subset Ethiopia, and compare the Ethiopia forecast against the Ethiopia July climatology (like with like).

```
MODELS = ["nmme/cansipsic4", "nmme/ccsm4", "nmme/cesm1"]

def nmme_july(y, region):
    fields = []
    for m in MODELS:
        d = rosetta.fetch(m, "precip", init=f"{y}-06", target=(datetime(y,7,1), datetime(y,7,31)),
                          hindcast=(y,y), region=region, verbose=False)["precip"]
        if d.sizes.get("init_time", 0) == 0:            # provider gap -> use the other models
            continue
        f = d.isel(init_time=0, drop=True)
        if float(f.mean()) > 20:                        # implausible July rate -> reject a corrupt model-
year
            continue
        fields.append(f)
    return xr.concat(xr.align(*fields, join="inner"), dim="model").mean("model")

train = range(2000, 2024)
hind_gha     = xr.concat([nmme_july(y, GHA) for y in train], dim=pd.Index(list(train), name="year"))
era_july_gha = era_gha.sel(time=era_gha.time.dt.month ==
7).groupby("time.year").mean("time").sel(year=list(train))

qm = get_method("qm")(); qm.fit(hind_gha, era_july_gha)
july_fc_gha = qm.predict(nmme_july(TARGET, GHA)).mean("member")
july_fc = july_fc_gha.sel(lat=slice(REGION[0], REGION[1]), lon=slice(REGION[2], REGION[3]))

eth_july_clim = (era.sel(time=era.time.dt.month == 7).groupby("time.year").mean("time")
                    .sel(year=list(train)).mean(["lat", "lon", "year"]))
print(f"bias-corrected July 2026 forecast (Ethiopia): {float(july_fc.mean()):.2f} mm/day "
      f"(Ethiopia July climatology {float(eth_july_clim):.2f} mm/day)")
```

```
bias-corrected July 2026 forecast (Ethiopia): 2.72 mm/day (Ethiopia July climatology 3.04 mm/day)
```

2 · Monitor the season to date (Stage 1)

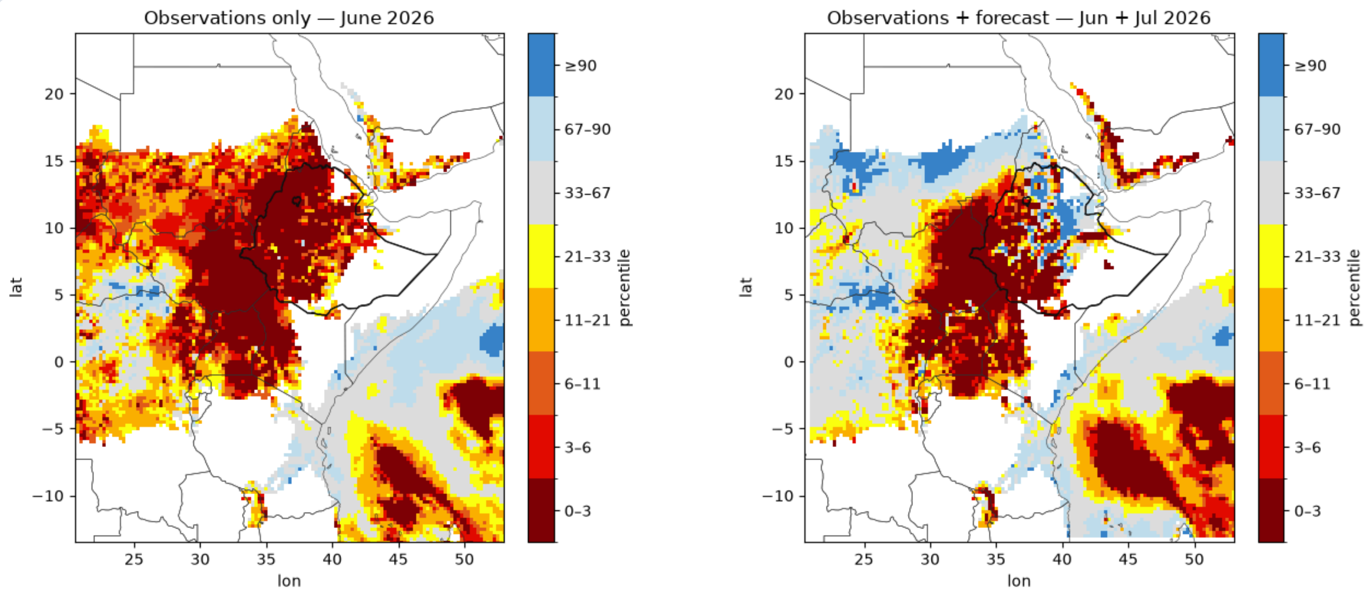
How the rainfall *so far* ranks against history, over the whole Greater Horn, two ways — observations only (June), and observations + the short-range forecast (June + first-half July).



CHC deck · slide 6 — CHIRPS season rank — observations only (left) and observations + forecast (right)

Both panels over the GHA extent, percentiles taken against the `CANDIDATES` climatology:

```
june = era_gha.sel(time=era_gha.time.dt.month == 6).groupby("time.year").mean("time")
july = era_gha.sel(time=era_gha.time.dt.month == 7).groupby("time.year").mean("time")

dry_mask = ((june + july) / 2).sel(year=CANDIDATES).mean("year") > 0.3
june_pct   = ds.percentile_of(june.sel(year=TARGET), june.sel(year=CANDIDATES)).where(dry_mask)
junjul_2026 = (june.sel(year=TARGET) + july_fc_gha) / 2
junjul_clim = ((june + july) / 2).sel(year=CANDIDATES)
junjul_2026, junjul_clim = xr.align(junjul_2026, junjul_clim, join="inner")
junjul_pct = ds.percentile_of(junjul_2026, junjul_clim).where(dry_mask)

borders = ds.natural_earth_borders(region=GHA)
eth_outline = woredas.dissolve()
fig, axes = plt.subplots(1, 2, figsize=(13.5, 5.6))
ds.plot_field_map(june_pct, ax=axes[0], classes=deck.PERCENTILE_CLASSES, boundaries=borders,
                  cbar_label="percentile", title="Observations only — June 2026")
ds.plot_field_map(junjul_pct, ax=axes[1], classes=deck.PERCENTILE_CLASSES, boundaries=borders,
                  cbar_label="percentile", title="Observations + forecast — Jun + Jul 2026")
for a in axes:
    eth_outline.boundary.plot(ax=a, color="#111", linewidth=1.0)
    a.set_xlim(GHA[2], GHA[3]); a.set_ylim(GHA[0], GHA[1])
fig.tight_layout(); plt.show()
```



The dry signal concentrates on the Ethiopian highlands (outlined), consistent with the deck. The observations+forecast panel is a little *less* extreme than the deck's, because our NMME July is near normal — the forecast substitute, not the observed record.

3 · The analog years (Stage 3a) — a forecast anchor, and the deck's curation

The analog selection needs a 2026 Kiremt-season Niño-3.4. Since only June 2026 is observed, the appropriate object is a **forecast** of the JJAS season — which is also what the deck's "rapidly intensifying El Niño" premise is. We compute it from the NMME models (June init), splice it onto the observed history, and select analogs as the **moderate-to-very-strong El Niño years** the deck curates:

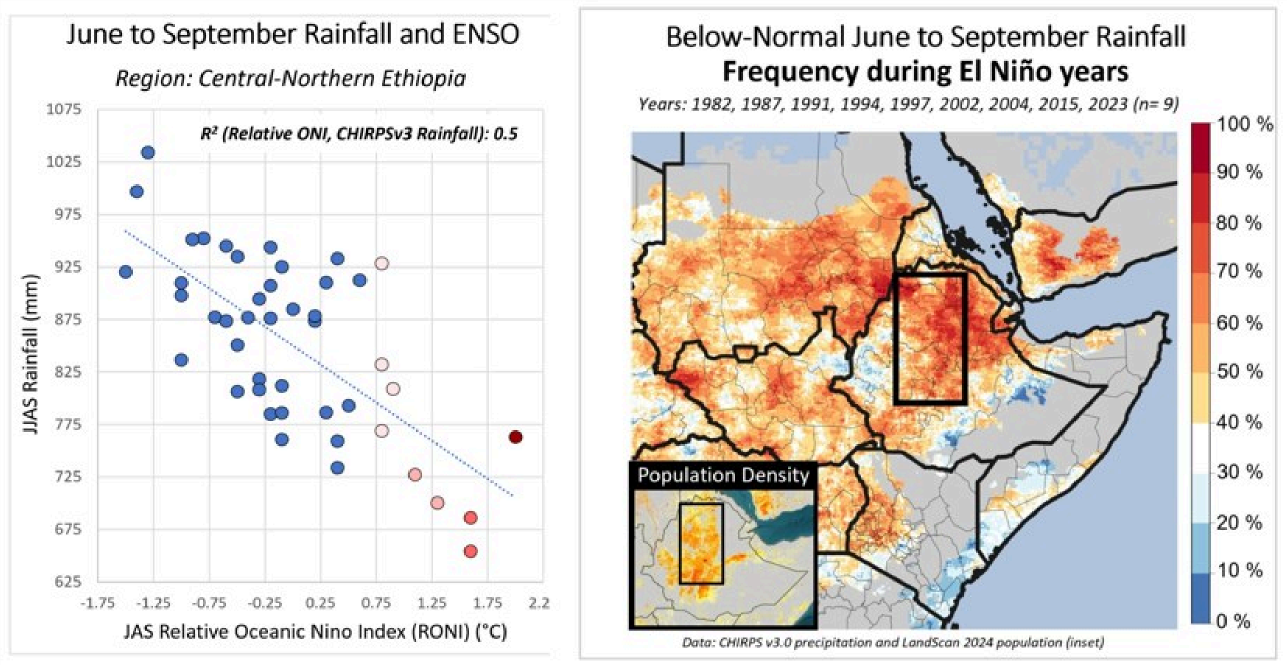
```python
sst = rosetta.fetch("obs/ersst-v5", "sst", hindcast=(1991, 2026), verbose=False)["sst"]

# Niño-3.4 as a RAW box-mean SST; we form the anomaly ourselves against the observed
# 1991-2020 climatology, so a single-year forecast field (no baseline of its own) can
# still be anomalised. (The forecast anchor is thus a perfect-prognosis estimate:
# model SST minus the OBSERVED climatology; any model mean-bias is folded in, which is
# acceptable for choosing *which* analog years 2026 resembles.)
n34_box = ds.Index.custom(name="n34", regions={"n34": [-5, 5, 190, 240]},
                          combine=lambda b: b["n34"], transform="raw", weights="cos_lat")

# observed JJAS Niño-3.4 (absolute) for COMPLETE seasons only (<= 2025), then anomalise
jjas_obs = (sst.sel(time=sst.time.dt.month.isin([6,7,8,9]) & (sst.time.dt.year <= 2025))
              .groupby("time.year").mean("time"))
n34_obs_abs = n34_box.reduce(jjas_obs)
clim = float(n34_obs_abs.sel(year=slice(BASELINE[0], BASELINE[1])).mean())
nino34_obs = n34_obs_abs - clim

# 2026 Kiremt-season Niño-3.4 as an NMME June-init FORECAST (not an observed month)
def nmme_jjas_sst(y):
    fields = []   # tropical Pacific band covers the Niño-3.4 box
    for m in MODELS:
        d = rosetta.fetch(m, "sst", init=f"{y}-06", target=(datetime(y,6,1), datetime(y,9,30)),
                          region=[-15,15,150,280], hindcast=(y,y), verbose=False)["sst"]
        if d.sizes.get("init_time", 0) == 0:
            continue
        fields.append(d.isel(init_time=0).mean("member"))
    return xr.concat(xr.align(*fields, join="inner"), dim="model").mean("model")
nino34_fc = float(n34_box.reduce(nmme_jjas_sst(TARGET).expand_dims(year=[TARGET])).sel(year=TARGET)) - clim

nino34 = xr.concat([nino34_obs, xr.DataArray([nino34_fc], dims="year", coords={"year":[TARGET]})],
dim="year")
print(f"2026 JJAS Niño-3.4 = {nino34_fc:+.2f} °C  (NMME June-init FORECAST anomaly vs 1991-2020 obs)")

# moderate-to-very-strong El Niño years = the deck's analog class
NINO_THR = 0.5                                          # JJAS developing-El-Niño threshold
cand = nino34.sel(year=CANDIDATES)
strong = sorted(int(y) for y, v in zip(cand.year.values, cand.values) if float(v) >= NINO_THR)
analogs = ds.analogs_from_index(nino34, target_year=TARGET, n=len(strong), candidates=strong)
print(f"moderate-to-very-strong El Niño analogs (JJAS Niño-3.4 >= {NINO_THR}):")
for y in strong:
    print(f"    {y}: {float(nino34.sel(year=y)):+.2f} °C")
```

```
2026 JJAS Niño-3.4 = +1.66 °C  (NMME June-init FORECAST anomaly vs 1991-2020 obs)
moderate-to-very-strong El Niño analogs (JJAS Niño-3.4 >= 0.5):
    1991: +0.59 °C
    1997: +1.66 °C
    2002: +0.85 °C
    2004: +0.54 °C
    2009: +0.51 °C
    2015: +1.69 °C
    2023: +1.20 °C
```

The anchor is a **forecast** of a developing El Niño, and the analog set is the **moderate-to-very-strong** El Niño years — matching the deck's stated curation. Note the outlook is *conditional* on that forecast El Niño developing, so its confidence tracks the ENSO forecast's own uncertainty.

4 · El-Niño rainfall relationship (Stage 3a diagnostic)

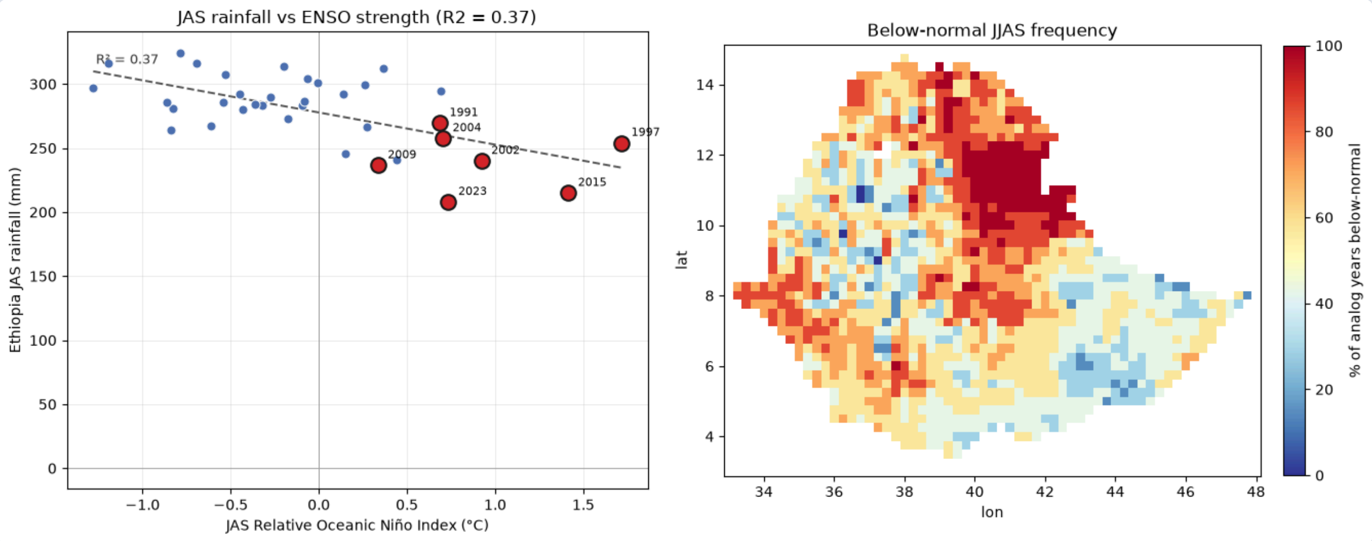
A scatter (stronger JAS **RONI** → lower Ethiopian JAS rainfall) beside the **composite map** — per cell, the share of analog years that were below-normal.



CHC deck · slide 8 — Slide 8 — JAS RONI vs rainfall (left) and the below-normal analog composite (right)

Both on one row, with the computed R^2:

```
jas_sst = sst.sel(time=sst.time.dt.month.isin([7,8,9])).groupby("time.year").mean("time")
roni_jas = ds.Index.named("roni", baseline=BASELINE).reduce(jas_sst)
jas_rain =
era.sel(time=era.time.dt.month.isin([7,8,9])).groupby("time.year").mean("time").mean(["lat","lon"]) * 92
jas_rain.name = "jas_rainfall_mm"
yrs = [y for y in CANDIDATES if y in roni_jas.year.values]
r2 = float(np.corrcoef(roni_jas.sel(year=yrs).values, jas_rain.sel(year=yrs).values)[0,1])**2

step_dims = [d for d in climatology.dims if d not in ("year", "lat", "lon")]
jjas_total = climatology.sum(step_dims)
below_freq = ds.frequency_below(jjas_total.sel(year=sorted(analogs.years.tolist())), jjas_total, q=1/3) * 100

fig, axes = plt.subplots(1, 2, figsize=(13.5, 5.4))
ds.plot_index_scatter(roni_jas.sel(year=yrs), jas_rain.sel(year=yrs), ax=axes[0],
        highlight=sorted(analogs.years.tolist()), highlight_color="#d62728", trendline=True,
        xlabel="JAS Relative Oceanic Niño Index (°C)", ylabel="Ethiopia JAS rainfall (mm)",
        title=f"JAS rainfall vs ENSO strength (R2 = {r2:.2f})")
ds.plot_field_map(below_freq, ax=axes[1], clip=woredas, cmap="RdYlBu_r", vmin=0, vmax=100,
        cbar_label="% of analog years below-normal", title="Below-normal JJAS frequency")
fig.tight_layout(); plt.show()
print(f"ENSO-rainfall fit: R2 = {r2:.2f}  (ERA5 from 1991; the deck reports ~0.5 on CHIRPS from 1981)")
print(f"region-mean below-normal frequency across analogs: {float(below_freq.mean()):.0f}% (coin-flip =
33%)")
```




```
ENSO-rainfall fit: R2 = 0.37  (ERA5 from 1991; the deck reports ~0.5 on CHIRPS from 1981)
region-mean below-normal frequency across analogs: 60% (coin-flip = 33%)
```

The negative slope reproduces the deck's — warmer JAS RONI, drier Ethiopian rains — with the analog years (red) at the warm, dry end. The R^2 is lower than the deck's ≈0.5, computed on the shorter ERA5 record from 1991.

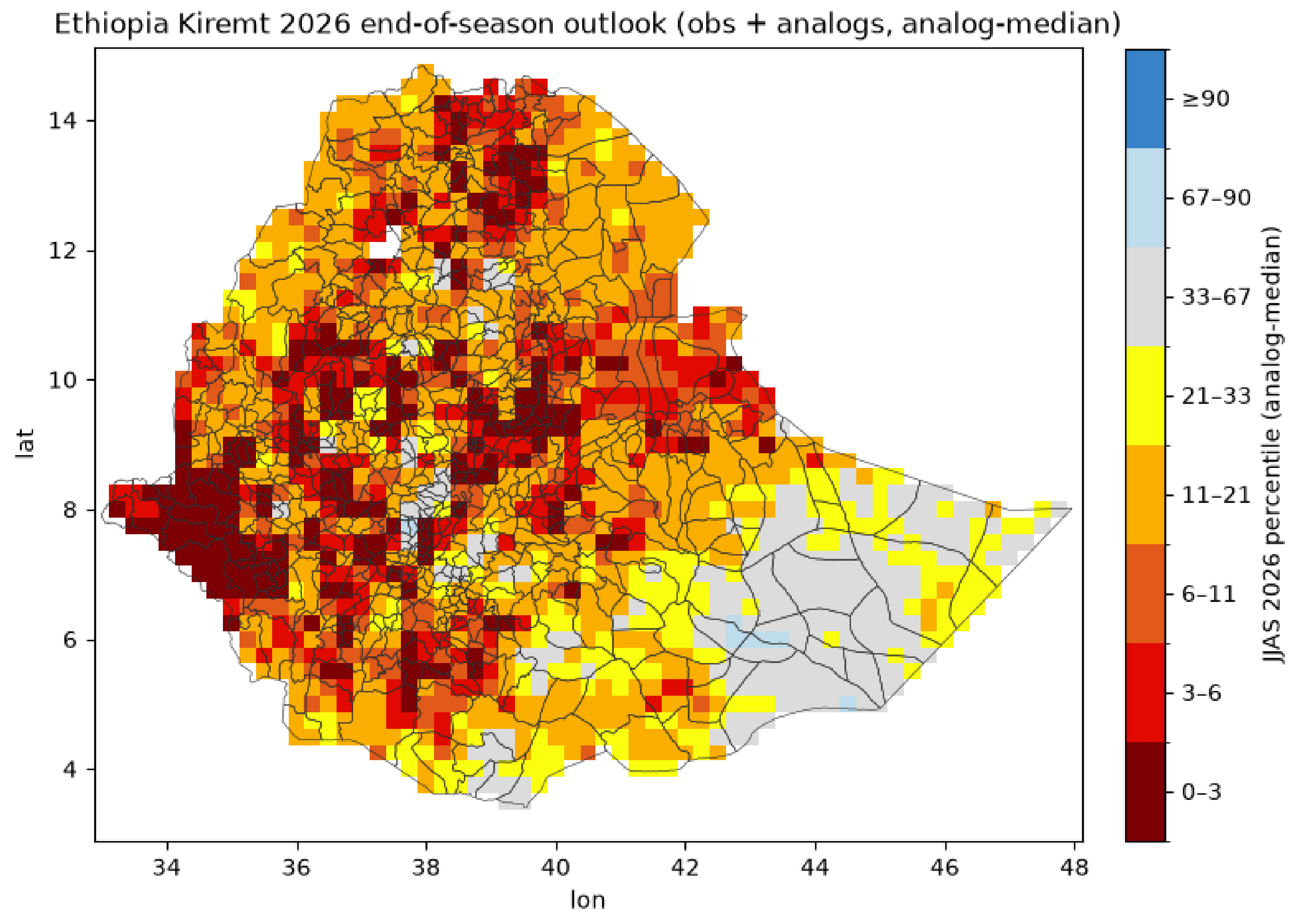
5 · Scenario completion (Stage 3)

`complete()` splices observed-to-date rainfall with each analog year's remainder, one end-of-season outcome per analog, and reads off the percentile:

```
obs = era.sel(time=(era.time.dt.year == TARGET) & (era.time.dt.month.isin([6,7,8,9])))
result = ds.complete(obs, analogs, climatology=climatology, season=SEASON, year=TARGET, cadence="monthly")
pct = result.percentile
print(f"end-of-season outlook: region-mean percentile {float(pct.mean()):.2f}; "
      f"{100*float((pct<0.21).mean()):.0f}% below 21st, {100*float((pct<0.33).mean()):.0f}% below 33rd")
```

```
end-of-season outlook: region-mean percentile 0.20; 68% below 21st, 80% below 33rd
```

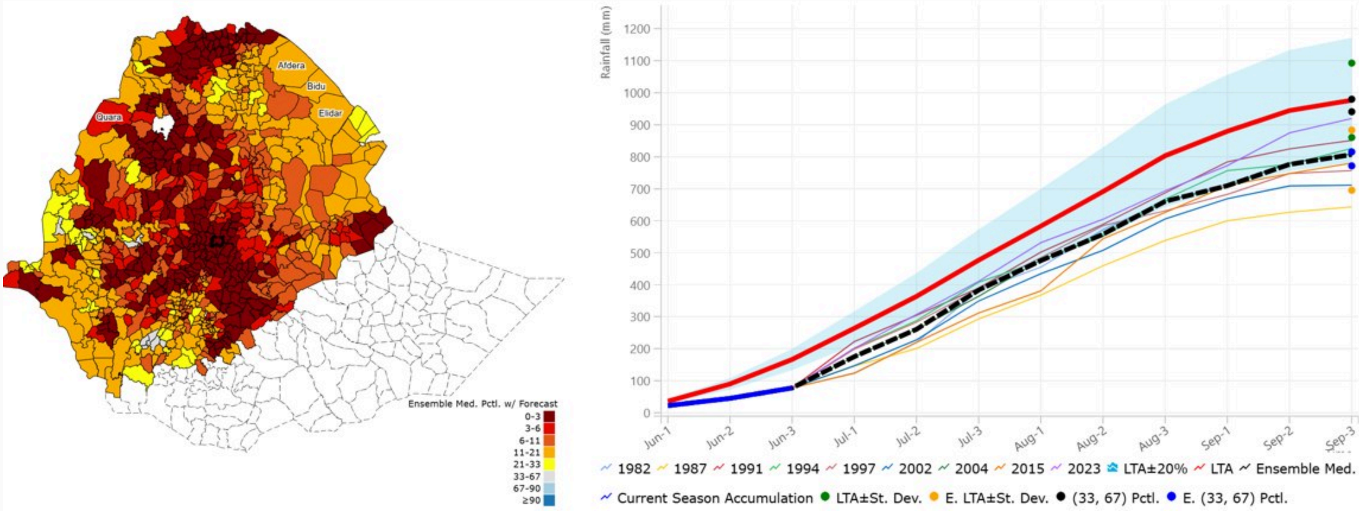
```
fig = ds.plot_field_map(pct, boundaries=woredas, clip=woredas, classes=deck.PERCENTILE_CLASSES,
                        cbar_label="JJAS 2026 percentile (analog-median)",
                        title="Ethiopia Kiremt 2026 end-of-season outlook (obs + analogs, analog-median)")
plt.show()
```



The analog-based outlook: a large share of the highlands below the 21st percentile. The map shows the **median** of the analog scenarios, so it is a *conditional analog-median* — a single number that does not convey the spread across the individual analog completions. This obs+analog configuration, using no dynamic forecast, is the part of the deck we match most directly.

6 · Woreda-level reporting and a district drill-down (Stage 3)

The SMPG reports at **woreda** level and pairs the choropleth with a single-district drill-down. The deck's illustrative district is **Sebeta Town** — a highland long-cycle-crop woreda just south-west of Addis.

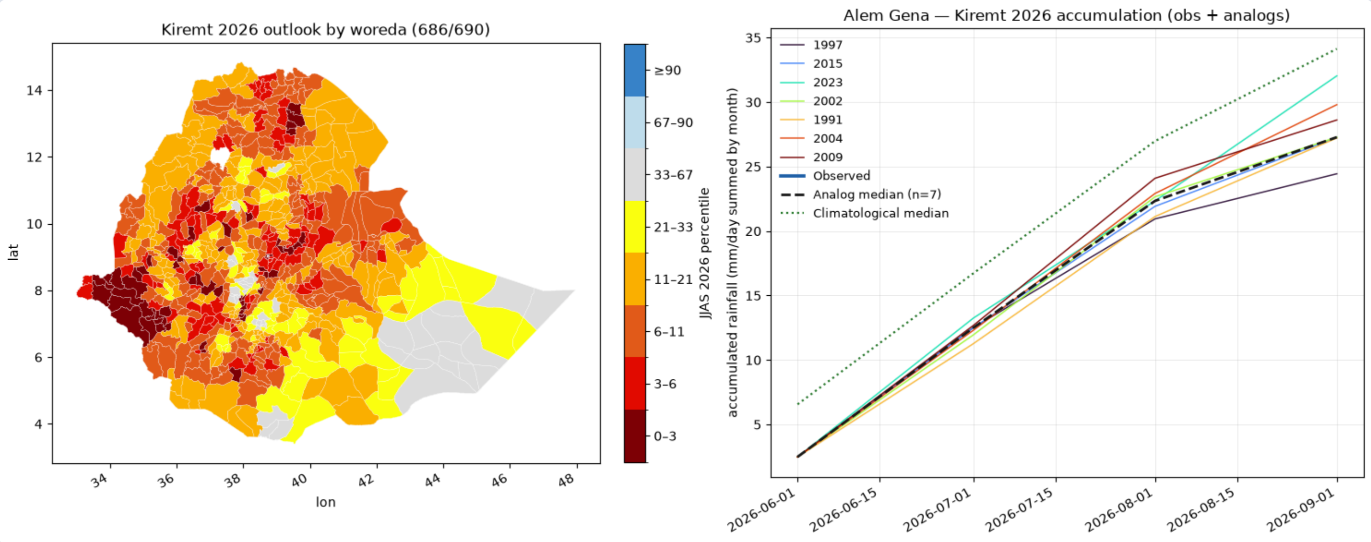


CHC deck · slide 12–13 — Slides 12–13 — the woreda percentile map (left) and a single-district accumulation (right)

`rosetta.zonal` reduces the percentile field over all 690 woredas. ERA5's 0.25° grid is coarser than many woredas, so we resample onto a fine mesh for coverage (this copies, it does not add detail). For the drill-down we select a **highland cropping district near Sebeta's location**, representative of the deck's example:

```python
def to_woreda(field):
    fine_lat = np.arange(float(field.lat.min()), float(field.lat.max()), 0.05)
    fine_lon = np.arange(float(field.lon.min()), float(field.lon.max()), 0.05)
    fine = field.interp(lat=fine_lat, lon=fine_lon, method="nearest")
    return rosetta.zonal(fine, woredas, by="shapeID", label="shapeName",
                         stat="mean", all_touched=False).rename("percentile")

per_woreda = to_woreda(pct)
resolved = int(np.isfinite(per_woreda.values).sum())

# highland cropping district near Sebeta Town (~8.92 N, 38.62 E)
sebeta = Point(38.62, 8.92)
woredas["_d"] = woredas.geometry.centroid.distance(sebeta)
sub = woredas.nsmallest(1, "_d"); wname = sub["shapeName"].iloc[0]
w_series = rosetta.zonal(era, sub, stat="mean", all_touched=True).isel(region=0)
w_clim = ds.seasonal_stack(w_series, SEASON, years=CANDIDATES)
w_obs  = w_series.sel(time=(w_series.time.dt.year == TARGET) & (w_series.time.dt.month.isin([6,7,8,9])))
res_w  = ds.complete(w_obs, analogs, climatology=w_clim, season=SEASON, year=TARGET, cadence="monthly")
print(f"{resolved}/690 woredas resolved | highland drill-down: {wname} "
      f"(percentile {float(res_w.percentile.mean()):.2f})")

fig, axes = plt.subplots(1, 2, figsize=(15, 6))
ds.plot_choropleth(per_woreda, woredas, by="shapeID", ax=axes[0], classes=deck.PERCENTILE_CLASSES,
                   cbar_label="JJAS 2026 percentile", title=f"Kiremt 2026 outlook by woreda
({resolved}/690)")
ds.plot_accumulation_scenarios(res_w, ax=axes[1], climatology=w_clim, color_by_scenario=True,
        ylabel="accumulated rainfall (mm/day summed by month)",
        title=f"{wname} — Kiremt 2026 accumulation (obs + analogs)")
fig.tight_layout(); plt.show()
```

```
686/690 woredas resolved | highland drill-down: Alem Gena (percentile 0.03)
```



The district outlook shows widespread below-normal rainfall in the cropping highlands, with the highland drill-down's analog scenarios fanning below its local climatological band — a representative case, not the single driest cell.

7 · Add the dynamic forecast (Stage 2)

Config B replaces the analog-filled July with the NMME forecast. This is where our substitute and the deck diverge:

```
july_fc_t = july_fc.expand_dims(time=[np.datetime64(f"{TARGET}-07-01")])
resB = ds.complete(obs, analogs, climatology=climatology, season=SEASON, year=TARGET,
                   cadence="monthly", forecast=july_fc_t, overlap="observed")
pctB = resB.percentile
print(f"config A (obs+analogs):          {100*float((pct<0.21).mean()):.0f}% below 21st,
{100*float((pct<0.33).mean()):.0f}% below 33rd")
print(f"config B (obs+forecast+analogs): {100*float((pctB<0.21).mean()):.0f}% below 21st,
{100*float((pctB<0.33).mean()):.0f}% below 33rd")
```

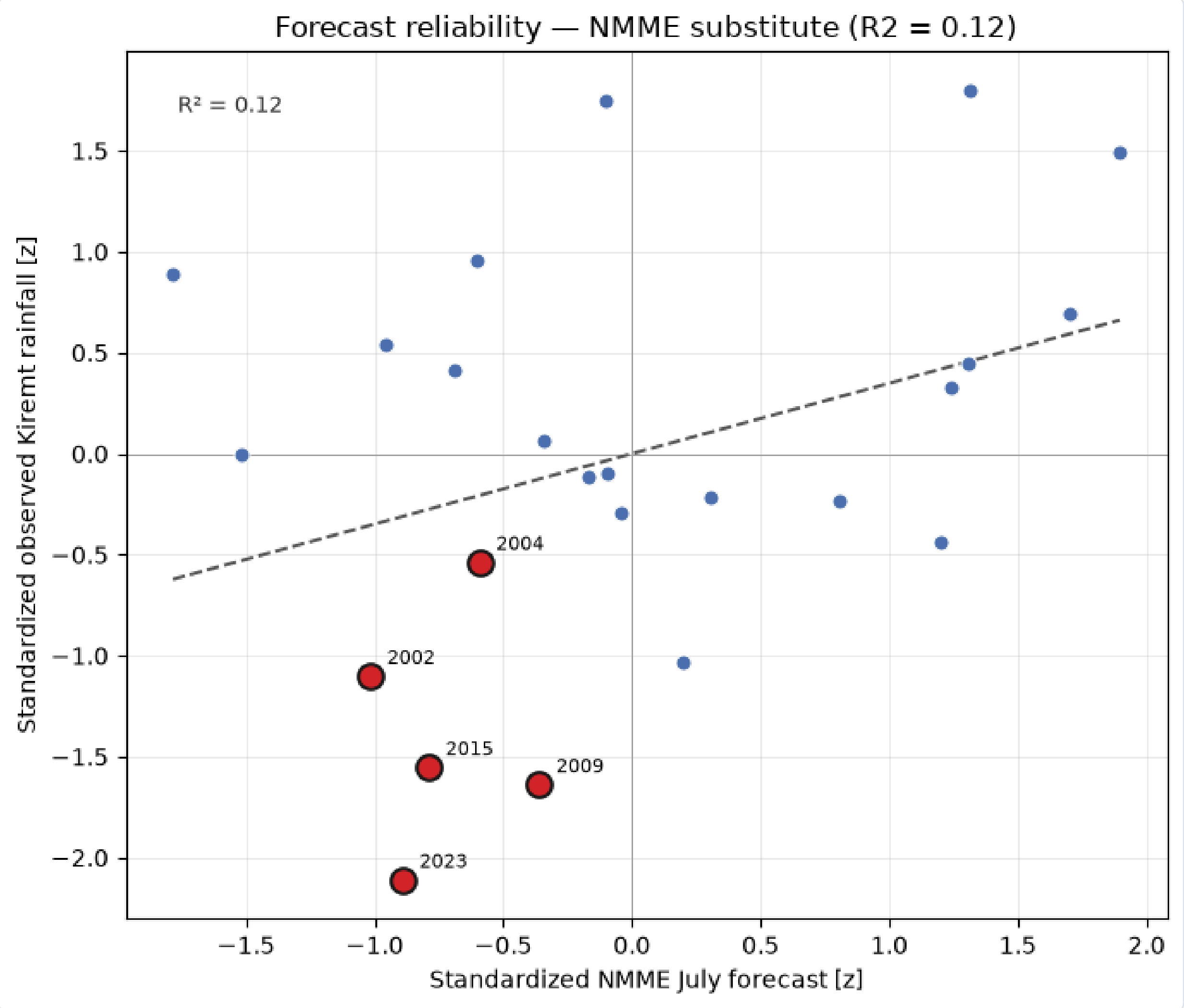
```
config A (obs+analogs):          68% below 21st, 80% below 33rd
config B (obs+forecast+analogs): 53% below 21st, 67% below 33rd
```

Our config B is *wetter* than config A (the near-climatology NMME July), whereas the deck's config B was *drier* (its CHIRPS-GEFS forecast an unusually dry July). So we do **not** reproduce the deck's headline intensification — the dynamic step softens here. The softening is the **forecast source**, not the ERA5 observations: because every percentile is taken within ERA5's own distribution, the ERA5↔CHIRPS mean bias (§10) cancels from the ranking.

8 · Should we trust the dynamic forecast? (Stage 4)

The deck's sub-seasonal ensemble scores $R^2 \approx 0.48$ for July; we verify the forecast we actually use — standardized NMME July vs standardized observed Kiremt rainfall:

```python
def zscore(a): return (a - a.mean()) / a.std()
fc_july    = hind_gha.sel(lat=slice(REGION[0], REGION[1]), lon=slice(REGION[2],
REGION[3])).mean("member").mean(["lat","lon"])
obs_kiremt =
(era.sel(time=era.time.dt.month.isin([6,7,8,9])).groupby("time.year").mean("time").mean(["lat","lon"]).sel(ye
ar=list(train)))
fx, fy = zscore(fc_july).rename("forecast_z"), zscore(obs_kiremt).rename("observed_z")
r2j = float(xr.corr(fx, fy))**2
dry_analogs = [y for y in sorted(analogs.years.tolist()) if y in list(train)]
fig = ds.plot_index_scatter(fx, fy, highlight=dry_analogs, highlight_color="#d62728", trendline=True,
        xlabel="Standardized NMME July forecast [z]", ylabel="Standardized observed Kiremt rainfall [z]",
        title=f"Forecast reliability — NMME substitute (R2 = {r2j:.2f})")
plt.show()
print(f"NMME July reliability R2 = {r2j:.2f}  (deck's sub-seasonal ensemble ~0.48)")
```

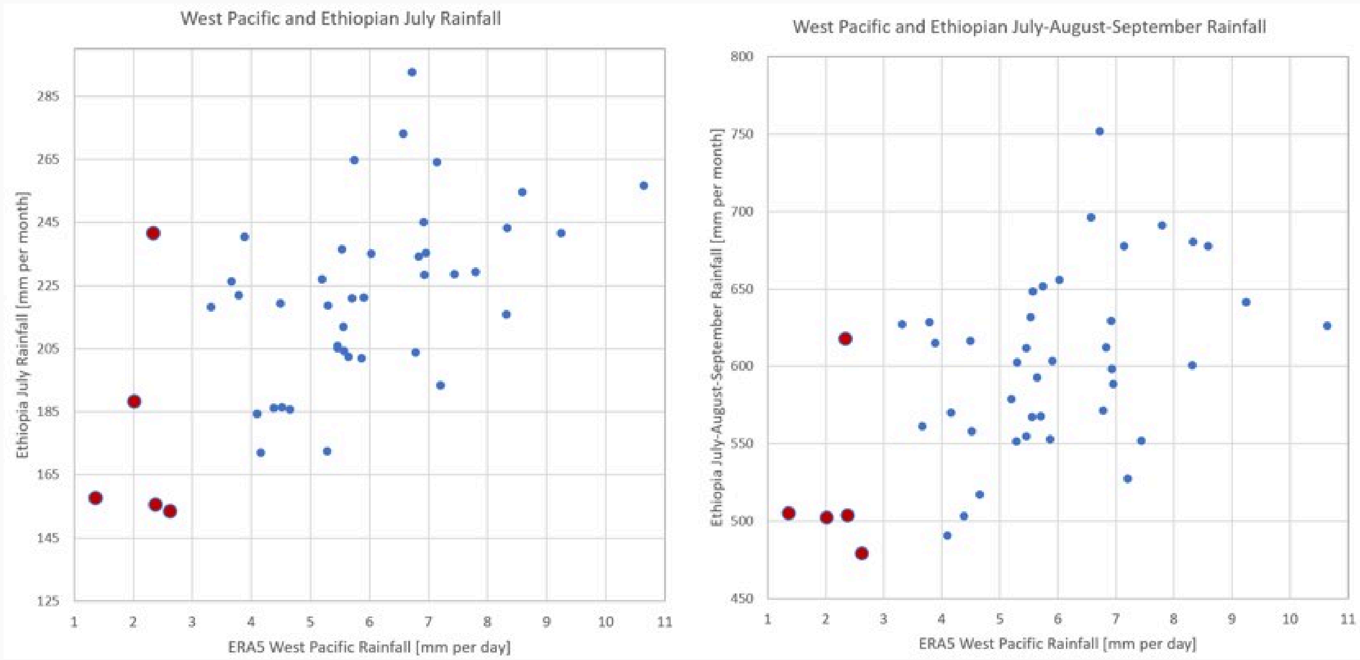



```
NMME July reliability R2 = 0.12  (deck's sub-seasonal ensemble ~0.48)
```

Our R^2 (~0.1) is **much lower** than the deck's ≈0.48: the slope is positive and the dry analogs still fall low-forecast/low-observed, but the seasonal NMME does not resolve the dry tail like the deck's tuned five-model sub-seasonal ensemble. This is the same weakness that softens config B. It motivates reconstructing the deck's West-Pacific leg from a different angle (§9b).

9 · The teleconnection cross-check (Stage 5, observational)

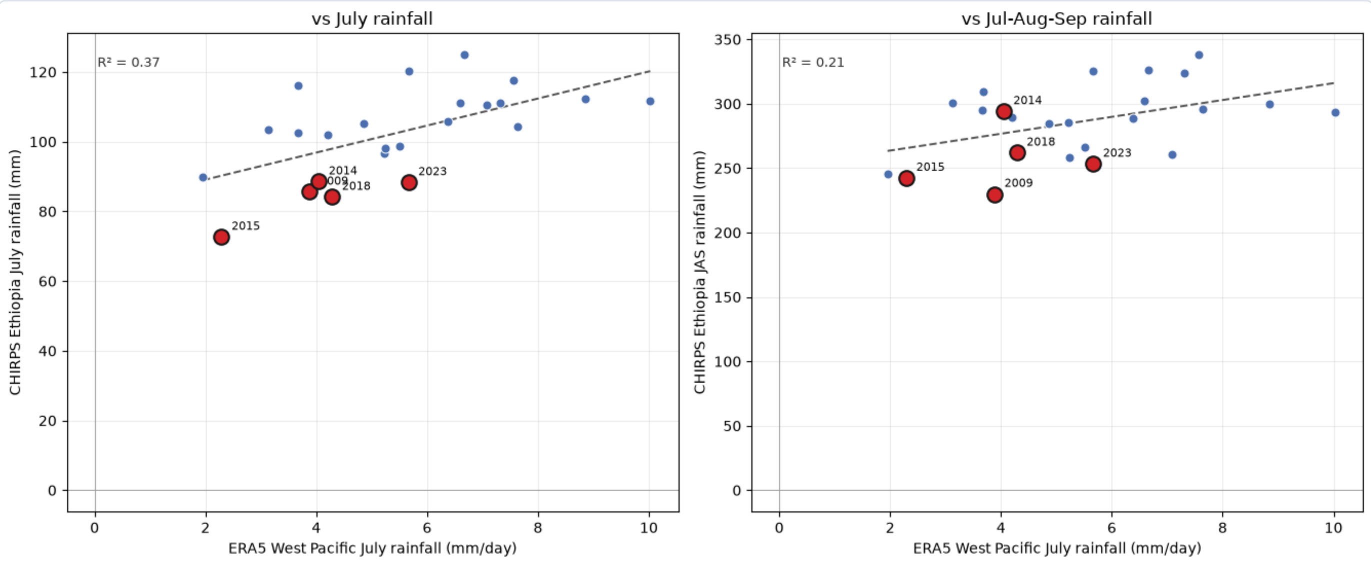
Dry **West Pacific** rainfall coincides with dry Ethiopian rainfall (weakened Walker circulation). The deck plots West-Pacific July rainfall against Ethiopian July (slide 19) and July–September (slide 20).



CHC deck · slide 19–20 — Slides 19–20 — West Pacific vs Ethiopian July (left) and July–September (right) rainfall

The deck's exact axes — **ERA5** West-Pacific vs **CHIRPS** Ethiopian rainfall (a backward-looking check, so CHIRPS's historical record suffices):

```
wpac = rosetta.fetch("obs/era5", "precip", hindcast=(2000, 2023), region=[-9, 4, 103, 140], verbose=False)
["precip"]
chirps = xr.concat([rosetta.fetch("obs/chirps-v3-daily-rhiza", "precip", hindcast=(y,y), region=REGION,
verbose=False)["precip"]
                    for y in range(2000, 2024)], dim="time").sortby("time")
wpac_jul = wpac.sel(time=wpac.time.dt.month == 7).groupby("time.year").mean("time").mean(["lat","lon"])
eth_jul  = chirps.sel(time=chirps.time.dt.month == 7).groupby("time.year").sum("time").mean(["lat","lon"])
eth_jas  =
chirps.sel(time=chirps.time.dt.month.isin([7,8,9])).groupby("time.year").sum("time").mean(["lat","lon"])
eth_jul, eth_jas = eth_jul.sel(year=wpac_jul.year), eth_jas.sel(year=wpac_jul.year)
wpac_jul.name = "wpac_july"; driest = eth_jul.sortby(eth_jul).year.values[:5].tolist()

fig, axes = plt.subplots(1, 2, figsize=(13.5, 5.6))
ds.plot_index_scatter(wpac_jul, eth_jul.rename("eth_july"), ax=axes[0], highlight=driest,
        highlight_color="#d62728", trendline=True, xlabel="ERA5 West Pacific July rainfall (mm/day)",
        ylabel="CHIRPS Ethiopia July rainfall (mm)", title="vs July rainfall")
ds.plot_index_scatter(wpac_jul, eth_jas.rename("eth_jas"), ax=axes[1], highlight=driest,
        highlight_color="#d62728", trendline=True, xlabel="ERA5 West Pacific July rainfall (mm/day)",
        ylabel="CHIRPS Ethiopia JAS rainfall (mm)", title="vs Jul-Aug-Sep rainfall")
fig.tight_layout(); plt.show()
```

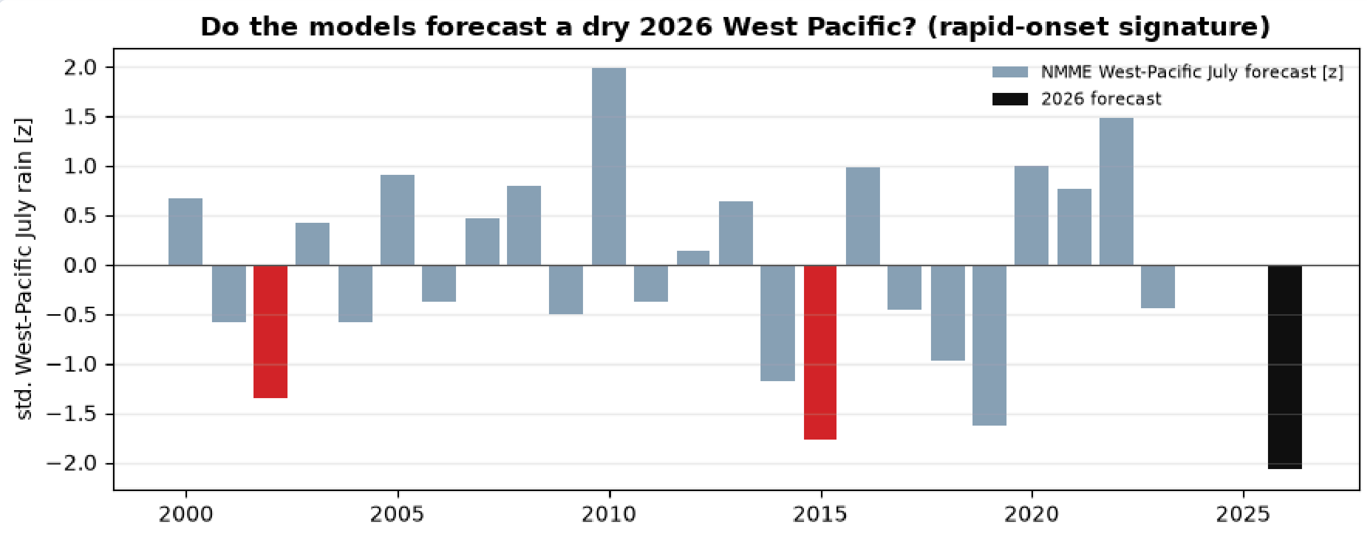


Both panels reproduce the deck's positive association — the driest Ethiopian years (red) at the dry end of the West-Pacific axis. This is the *observed* relationship (deck slides 19–20). The deck's **forecast** leg (slide 18) — do the models predict a dry 2026 West Pacific? — is reconstructed next.

9b · The West-Pacific forecast leg (the deck's slide 18)

The deck's fourth leg is a forecast argument: a **rapid El Niño onset drives a dry West Pacific**, as in 2002 and 2015, which weakens the Walker circulation and suppresses Kiremt. We build the **NMME** West-Pacific (9°S–4°N, 103–140°E) July precipitation forecast, standardized, and place 2026 against the history. This is not the deck's five sub-seasonal models, so it is a weaker instrument, used to check the *direction* of the signal:

```python
WPAC = [-9, 4, 103, 140]
def nmme_wpac_july(y):
    vals = []
    for m in MODELS:
        d = rosetta.fetch(m, "precip", init=f"{y}-06", target=(datetime(y,7,1), datetime(y,7,31)),
                          hindcast=(y,y), region=WPAC, verbose=False)["precip"]
        if d.sizes.get("init_time", 0) == 0:
            continue
        v = float(d.isel(init_time=0, drop=True).mean("member").mean(["lat","lon"]))
        if v > 40:                                       # implausible rate -> reject a corrupt model-year
            continue
        vals.append(v)
    return float(np.mean(vals))

wp_years = list(range(2000, 2024)) + [TARGET]
wp = pd.Series({y: nmme_wpac_july(y) for y in wp_years})
wp_z = (wp - wp.loc[2000:2023].mean()) / wp.loc[2000:2023].std()

fig, ax = plt.subplots(figsize=(9, 3.6))
hist = wp_z.loc[2000:2023]
ax.bar(hist.index, hist.values, color="#8aa0b8", label="NMME West-Pacific July forecast [z]")
for yr in [2002, 2015]:
    if yr in hist.index: ax.bar([yr], [hist.loc[yr]], color="#d62728")
ax.bar([TARGET], [wp_z.loc[TARGET]], color="#111", label="2026 forecast")
ax.axhline(0, color="#555", lw=0.8); ax.set_ylabel("std. West-Pacific July rain [z]")
ax.set_title("Do the models forecast a dry 2026 West Pacific? (rapid-onset signature)", fontweight="bold")
ax.legend(fontsize=8, frameon=False); ax.grid(alpha=0.25, axis="y"); fig.tight_layout(); plt.show()
print(f"2026 West-Pacific July forecast: {wp_z.loc[TARGET]:+.2f} z   "
      f"(dry-onset analogs 2002={wp_z.get(2002, float('nan')):+.2f}, 2015={wp_z.get(2015,
float('nan')):+.2f})")
```




```
2026 West-Pacific July forecast: -2.08 z   (dry-onset analogs 2002=-1.35, 2015=-1.77)
```

The sign of the 2026 forecast relative to the 2002/2015 rapid-onset years is the reconstructed leg: a **negative** (dry) 2026 West Pacific would corroborate the deck's rapid-onset argument from an independent forecast, a positive one would not. Read it as *directional* corroboration only — the three-model seasonal NMME is not the deck's dry-tail-tuned five-model sub-seasonal ensemble (§8), so it is a weaker instrument than the deck's, and this leg should be weighted accordingly.

10 · Is ERA5 a defensible stand-in for CHIRPS?

Measured over 2000–2023, where both exist:

```python
def jjas_total(da, per="sum"):
    s = da.sel(time=da.time.dt.month.isin([6,7,8,9])).groupby("time.year")
    return (s.sum("time") if per=="sum" else s.mean("time")*122).mean(["lat","lon"])
era_t, chp_t = xr.align(jjas_total(era, "mean"), jjas_total(chirps, "sum"), join="inner")
print(f"ERA5 vs CHIRPS JJAS totals, 2000–2023: Pearson r = {float(xr.corr(era_t, chp_t)):.2f}, "
      f"mean bias {float((era_t-chp_t).mean()):+.0f} mm")
```

```
ERA5 vs CHIRPS JJAS totals, 2000–2023: Pearson r = 0.81, mean bias −13 mm
```

ERA5 and CHIRPS track at **r ≈ 0.8** with a small dry bias. Because 2026 is positioned within ERA5's *own* distribution, that mean bias largely cancels from the percentiles. One caveat: this correspondence is a **seasonal-total, region-mean** check — it does not certify ERA5's *pixel-level tail* behaviour over the steep highlands, so the deepest "driest-on-record" pixels (especially in the wet south-west) should be read cautiously.

Recap

Three `fetch` calls (ERA5, ERSSTv5, NMME) and a sequence of DeepScale operations rebuild the SMPG stages for 2026: the observed-to-date monitoring (§2), an ENSO-forecast-anchored moderate-to-very-strong analog set (§3), the ENSO–rainfall relationship and below-normal composite (§4), the scenario completion at pixel, region, and woreda level (§5–6), the dynamic-forecast configuration (§7) and its reliability check (§8), and the West-Pacific teleconnection — observed (§9) and forecast (§9b).

The analog-driven outlook — the part independent of the substituted forecast — matches the deck closely: a widespread poor Kiremt, with the West-Pacific forecast pointing the same way. The one leg that diverges is the dynamic forecast: the NMME July is near climatology and softens the outlook, where the deck's drier CHIRPS-GEFS sharpened it. The outlook is conditional on the forecast El Niño developing, so its confidence tracks that of the ENSO forecast.